

Scenario 4:

SpeedCart Online

*By: Victoria Skeba, Tyler
Coulter, Gehrig Eckoff*



Background Information

- SpeedCart Online is a fast-growing U.S. e-commerce retailer, currently operating with a single fulfillment center in Kansas City, MO.
- We needed to evaluate the rapid growth in online orders that were straining the existing facility, which was also leading to delays in deliveries especially to distant regions.

Supply Center: Kansas City, MO | (2136, 711) km | Current central fulfillment center Capacity 700 units/week

Distribution Centers

Location	Coordinates	Fixed Costs	Capacity
Newark NJ	(3896, 892) km	\$130k	500 units/week
Atlanta GA	(3015, 117) km	\$110k	500 units/week
Dallas TX	(1945, 10) km	\$110k	500 units/week
Los Angeles CA	(95, 151) km	\$130k	500 units/week

Demand Center (Volume)

Location	Coordinates	Demand Low	Demand High
New York NY	(3911, 890)km	90	130
Chicago IL	(2736, 1020)km	80	120
Atlanta GA	(3015, 117)km	70	100
Dallas TX	(1945, 10) km	60	90
Los Angeles CA	(95, 151) km	100	150

Cost Analysis

Constraints:

- Each center has fixed setup costs and weekly capacity limits
- Transportation costs are \$0.70 per unit per kilometer
- Kansas City is already operational (capacity: 700 units/week)

Demand Uncertainty:

- 70% probability of a high-growth scenario (higher weekly demand)
- 30% probability of a steady (low-demand) scenario

Strategy A: Only East and West Coast Centers

Transportation Cost \$.70 per unit per km

Supply Center: Kansas City Fixed Cost \$0 (700 unit/wk capacity)

DC: Newark Fixed Cost \$130,000 (500 unit/wk capacity)

Los Angeles Fixed Cost \$130,000 (500 unit/wk capacity)

Strategy B: Open all four regional centers

Transportation Cost \$.70 per unit per km

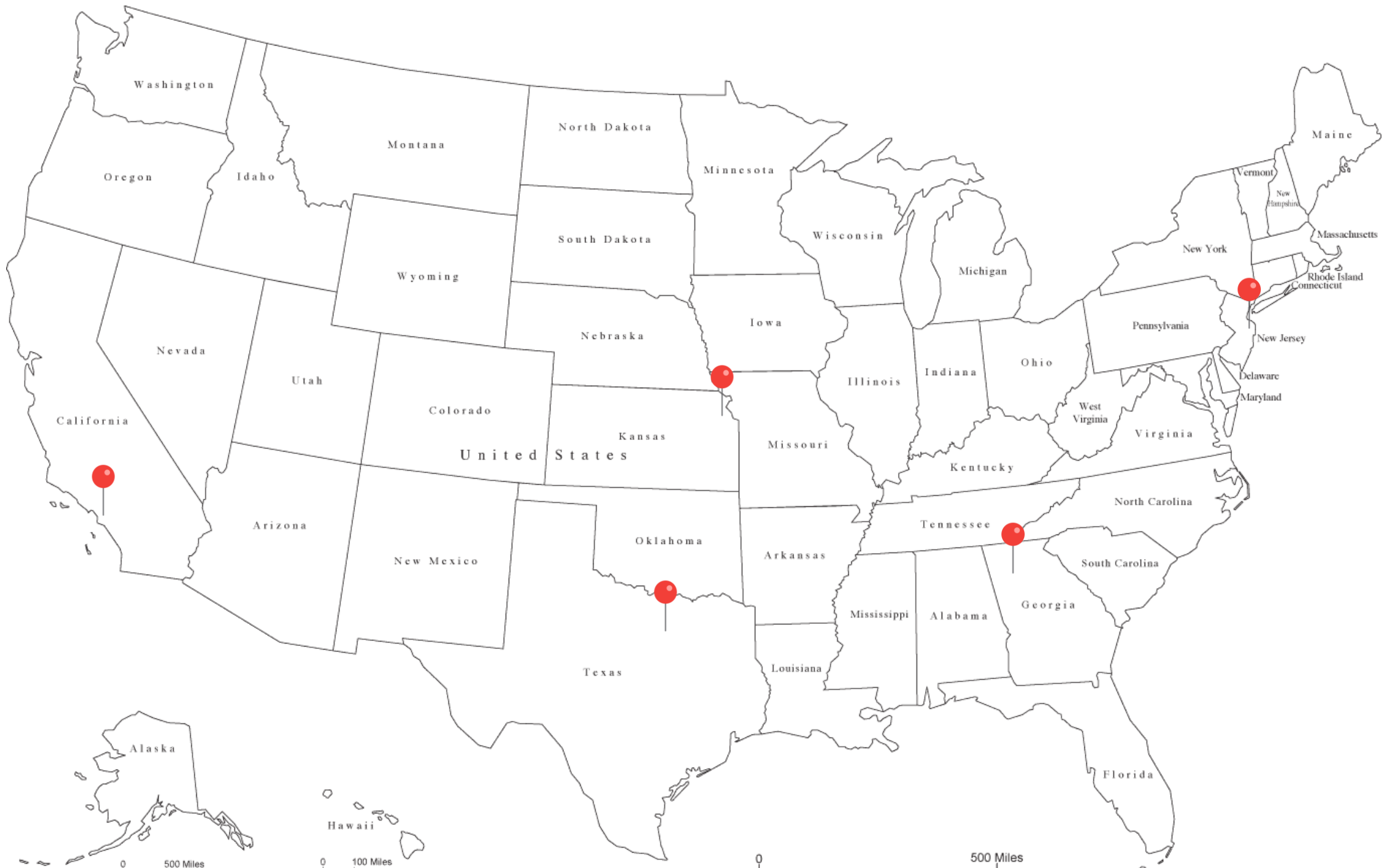
Supply Center: Kansas City Fixed Cost \$0 (700 unit/wk capacity)

DC: Atlanta Fixed Cost \$110,000 (500 unit/wk capacity)

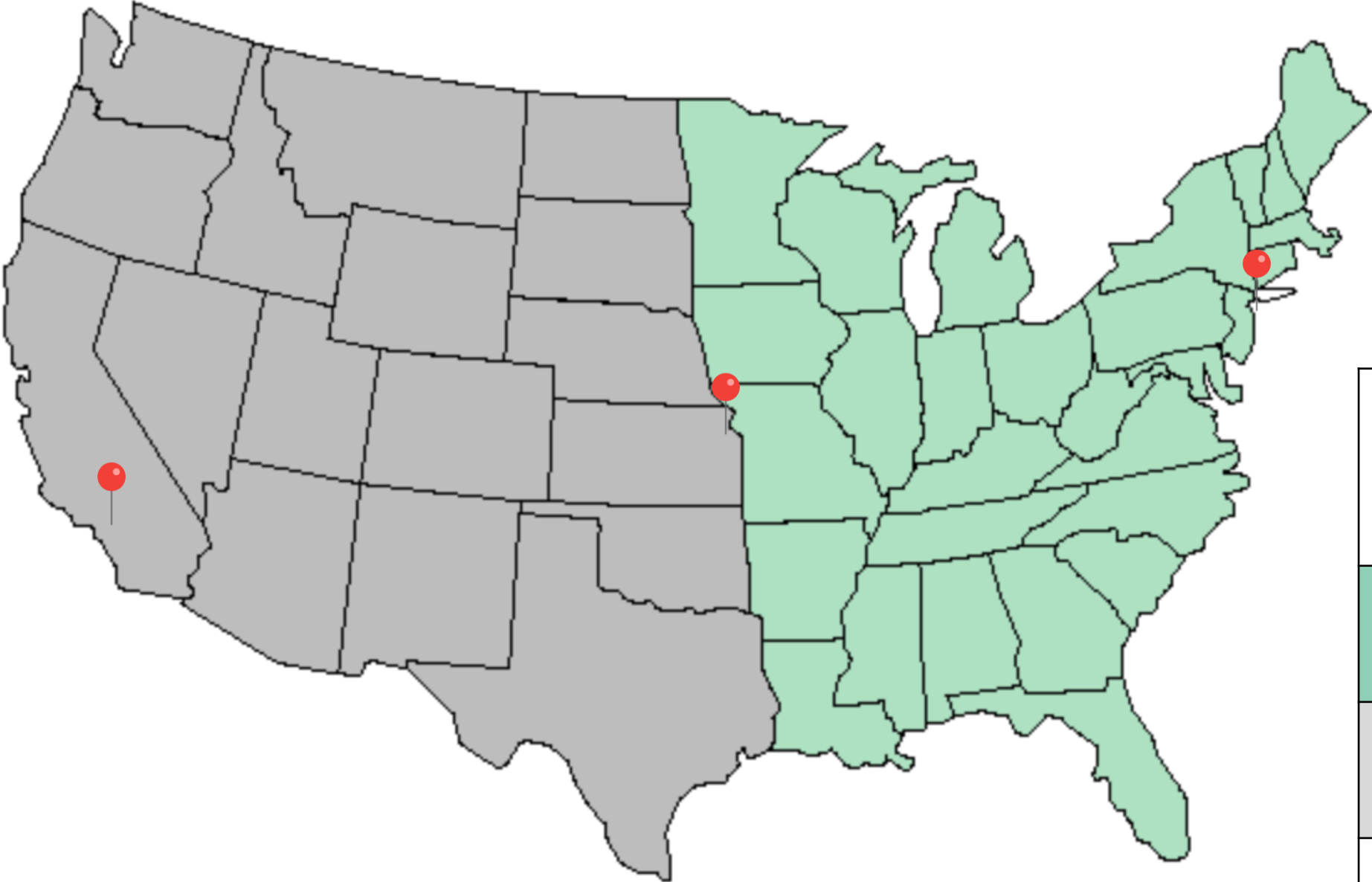
Dallas Fixed Cost \$110,000 (500 unit/wk capacity)

Newark Fixed Cost \$130,000 (500 unit/wk capacity)

Los Angeles Fixed Cost \$130,000 (500 unit/wk capacity)

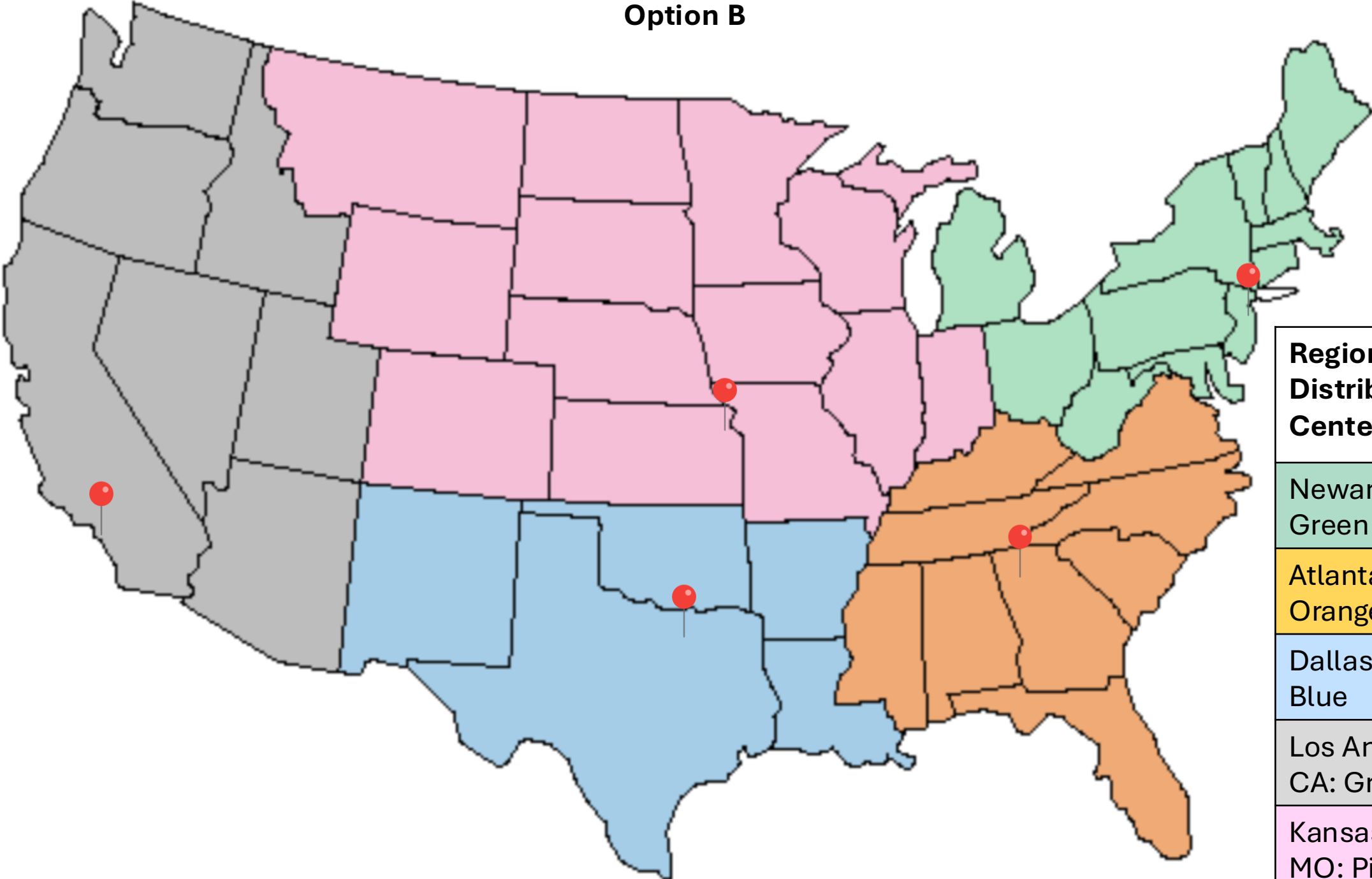


Option A



Regions for Distribution Centers
Newark, NJ: Green
Los Angeles: Grey
Kansas City, MO

Option B



Regions for Distribution Centers
Newark, NJ: Green
Atlanta, GA: Orange
Dallas, TX: Blue
Los Angeles, CA: Grey
Kansas City, MO: Pink

Total Annual Cost Calculations

Option	Distribution Centers	Initial Cost	High Demand Costs	Expected Cost
A	East & West Coast	\$41 M	\$69 M	\$59.6 M
B	All 4 Locations	\$48 M	\$61 M	\$56.1 M

